

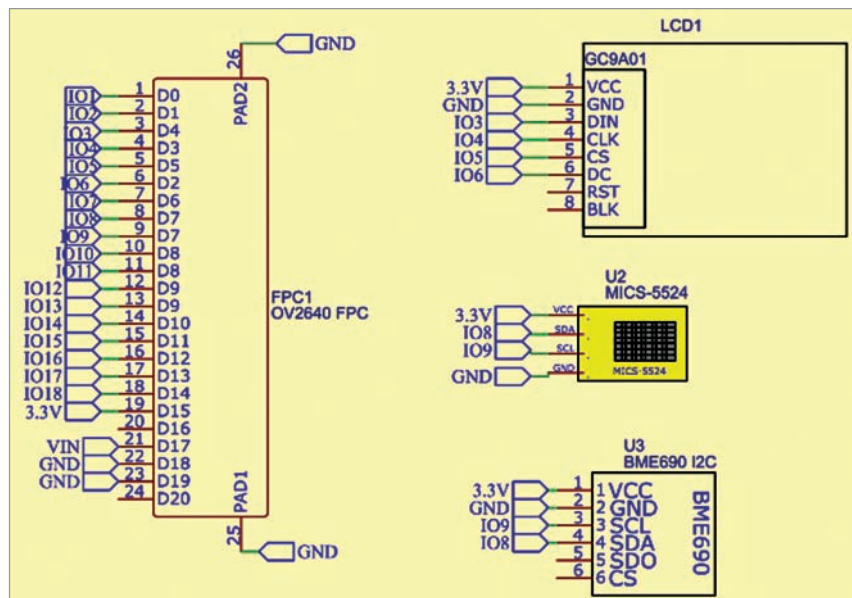


Fig. 5: FPC gas and environmental sensor module circuit diagram

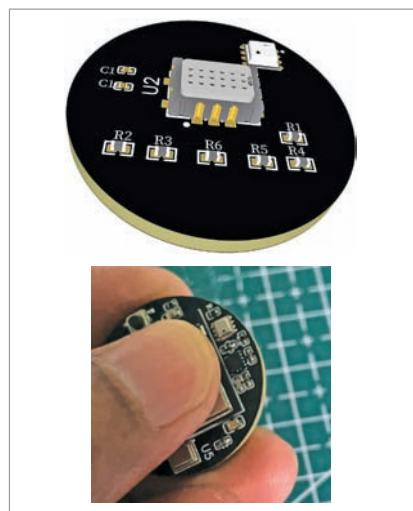


Fig. 6: PCB design of the sensor



Fig. 7: Sensor FPC connector

IndusBoard pins 8 and 9), with power supplied via the IndusBoard's 3.3V pin. The LCD connects via SPI, using pins 3, 4, 5, and 6. The detailed customised sensor module design will be covered in the next part of this series.

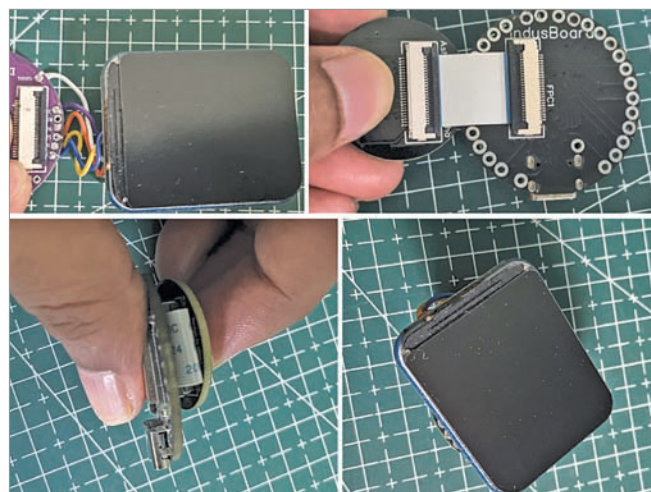


Fig. 8: Complete prototype

PCB design

The FPC gas and environmental sensor PCB is designed with solder pads on the bottom for connecting the display. All components connect to the IndusBoard Coin V2 via an FPC connector. The sensors and main

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components are placed on the top layer, while the bottom layer houses the FPC connector and SMD solder pads for the display, as shown in Fig. 6.

Construction and testing

First, upload the source code to the IndusBoard. Next, connect the small-sized 24-pin, 0.5mm FPC ribbon cable to the sensor module. The FPC connector is located on the back of the module, as shown in Fig. 7. Attach the FPC connector to the IndusBoard Coin V2. Solder the display wires to the SMD solder pads on the bottom of the sensor module. Then, connect the FPC connector to both the IndusBoard Coin and the sensor module. Finally, place the sensor module behind the IndusBoard Coin and the display on top of the Coin (see Fig. 8).

For testing, connect a 3.3V or 3.7V battery to the device. The device will begin displaying gas readings on the OLED display. Using the SSID and password set in the code earlier, connect a laptop, smart TV, or smartphone to the device's Wi-Fi access point (AP). Open a web browser and navigate to 192.168.4.1 to access the wireless IoT dashboard, which displays all gas and environmental sensor data in real time. The dashboard also provides buttons to record the data and save it as a CSV file. **EFY**

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